Sports Injury Risk Detection from Video

AI-Powered Biomechanical Analysis Platform for Athletic Injury Prevention

Presented By: Rachit Patnaik

Challenges Facing Sports Injury Management

Reactive Treatments

Injuries are traditionally addressed only after they occur — when it's already too late.

Subtle Movement Faults

Dangerous compensations are incredibly hard to catch with the naked eye during live performance.

Manual Processes

Current biomechanical assessments rely heavily on manual observation, introducing error and delay.

Inefficient Collaboration

Disconnect between athletes, coaches, and physiotherapists diminishes prevention effectiveness.

Project Objectives

1 Build an AI-Powered Platform

Analyze athlete movement videos to identify biomechanical issues in real time.

2 Assess Risk Factors

Predict potential injuries — ACL, hamstring, and more — before they occur.

3 Proactive Prevention

Enable continuous movement monitoring and ongoing risk assessment.

4 Actionable Guidance

Provide automated, personalized corrective exercise recommendations.

Core Features of the Platform

Biomechanical Pose Estimation

Automatic human pose detection and joint tracking using AI-powered computer vision.

Multi-Modal Sensor Fusion

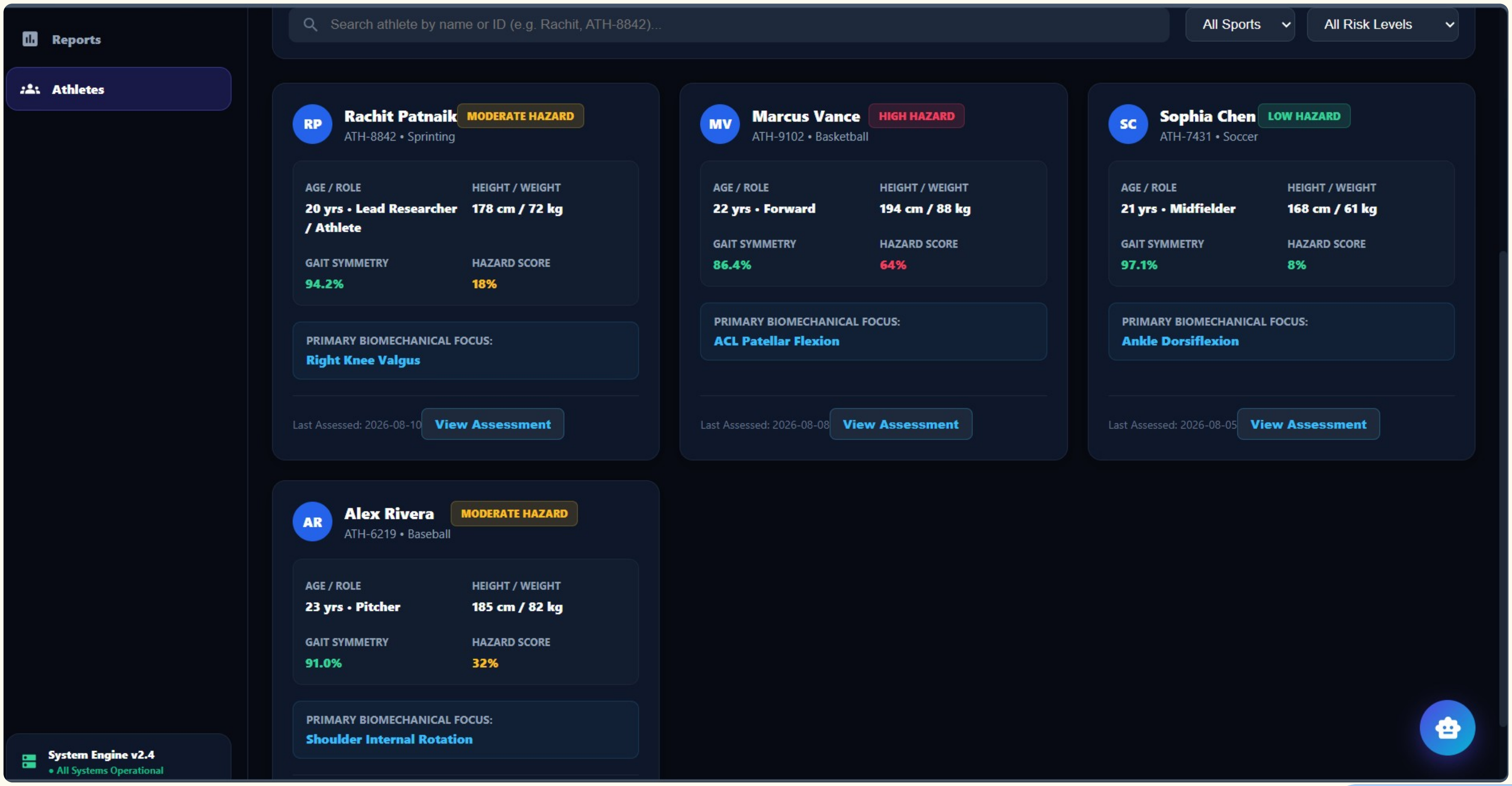
Synchronized telemetry overlaying joint flexion velocity against IMU impact data.

Multi-Role Dashboards

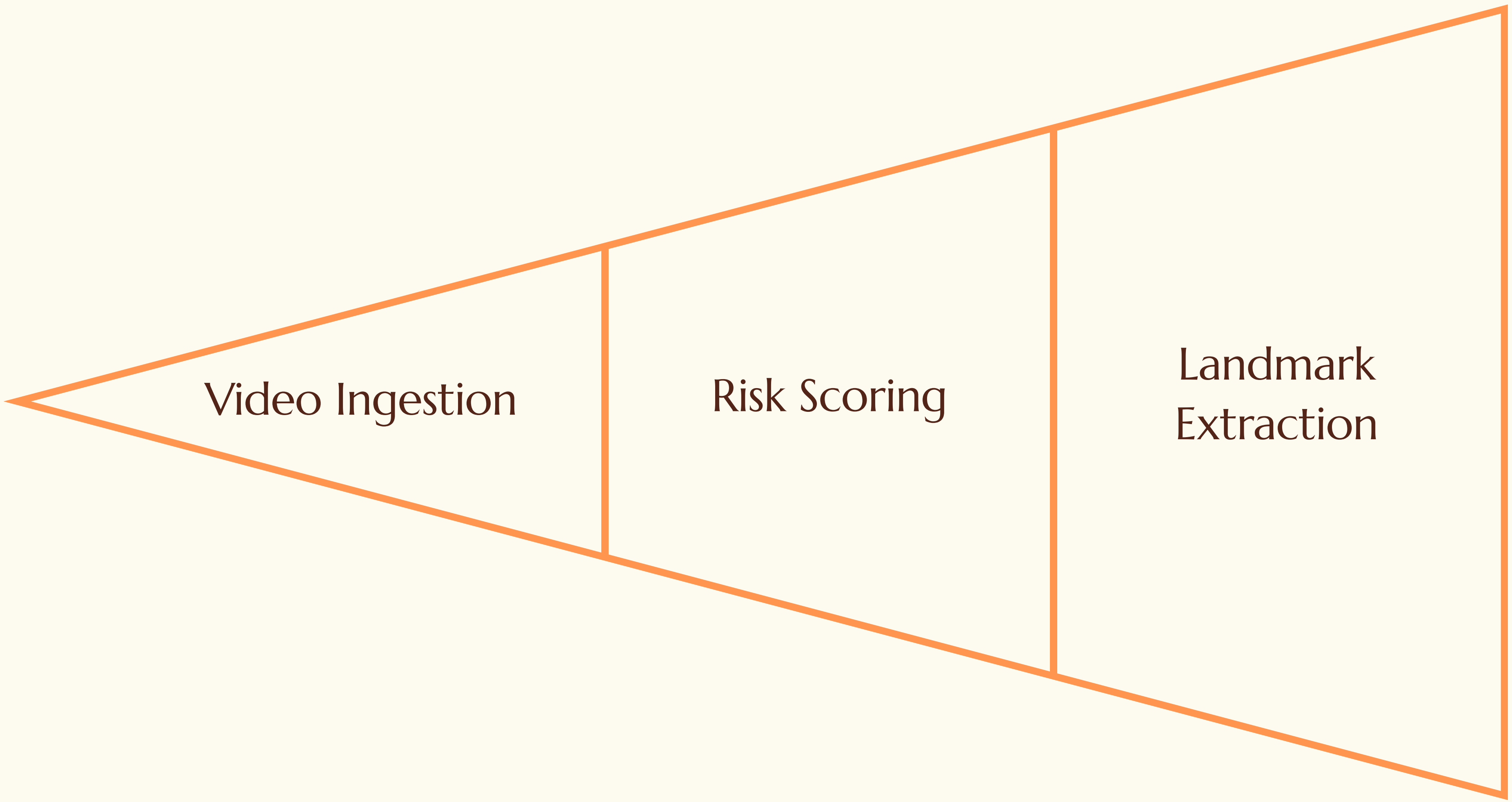
Secure interfaces for Athletes, Coaches, Physiotherapists, Scientists, and Admins.

Automated Clinical Reports

Printable PDF assessment summaries with QR verification for clinical use.



Methodology & Implementation



Each stage of the pipeline is designed to be modular and independently scalable, ensuring robust performance from raw video input through to actionable clinical output.

System Architecture Overview

Modular Microservices Layer

Video processing, pose estimation, and risk scoring operate independently for maximum resilience.

API Gateway (FastAPI)

Handles routing, request validation, and acts as the central execution entry point.

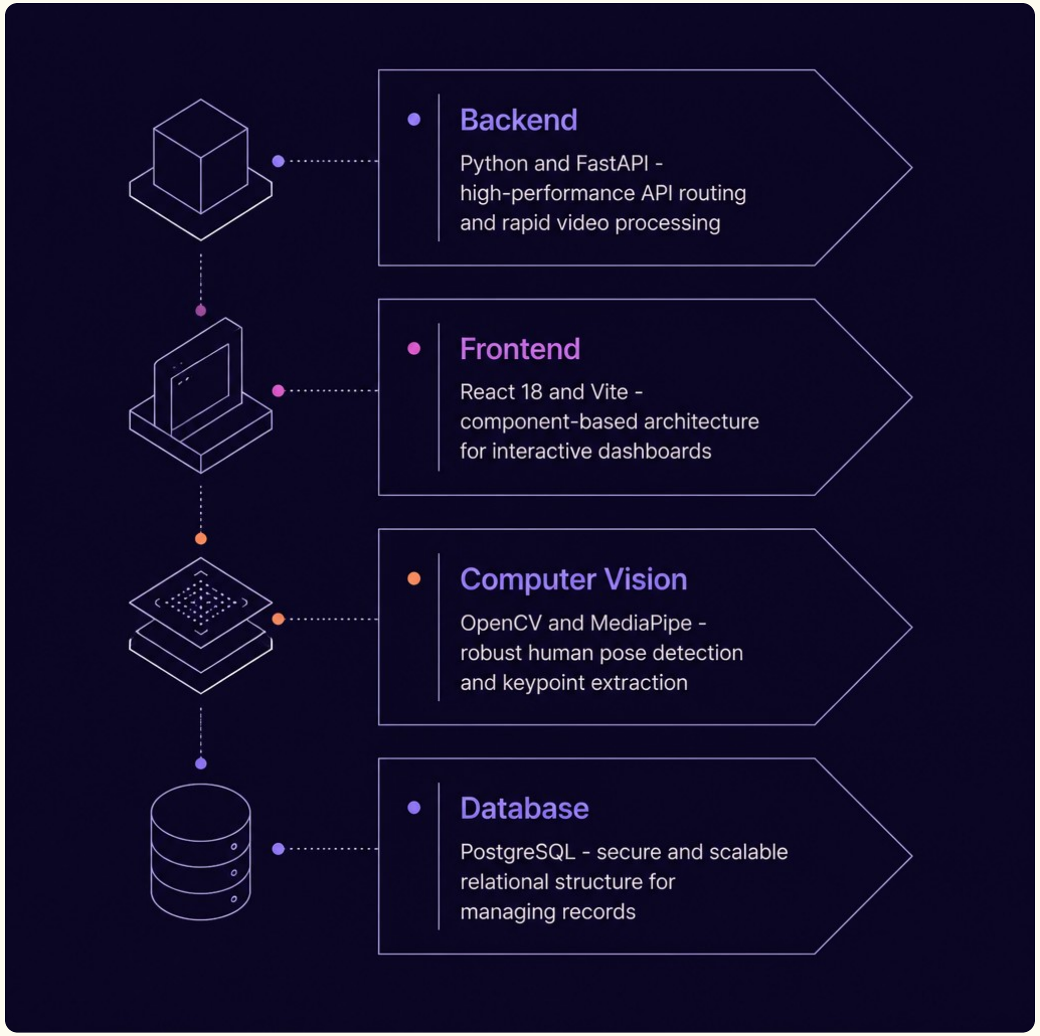
Data Security

JWT authentication and role-based access control (RBAC) safeguards all sensitive athlete data.

AI/ML Intelligence Layer

Deep learning models feed into time-series analysis pipelines for predictive analytics.

Technology Stack Explained



Backend

Python & FastAPI — High-performance API routing and rapid video processing pipeline execution.

Frontend

React 18 & Vite — Component-based architecture powering interactive, real-time dashboards.

Computer Vision

OpenCV & MediaPipe — Robust human pose detection and 33-keypoint extraction from video frames.

Database

PostgreSQL — Secure and scalable relational structure for managing all athlete records.

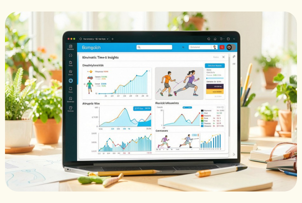
User Interface & Mock-ups

Real-Time Live WebCam Screening

Live pose tracking calculating dynamic joint angles at 30 FPS.

Coach Analytics Dashboard

Centralized coaching view for reviewing athlete performance, trends, and session-level insights.

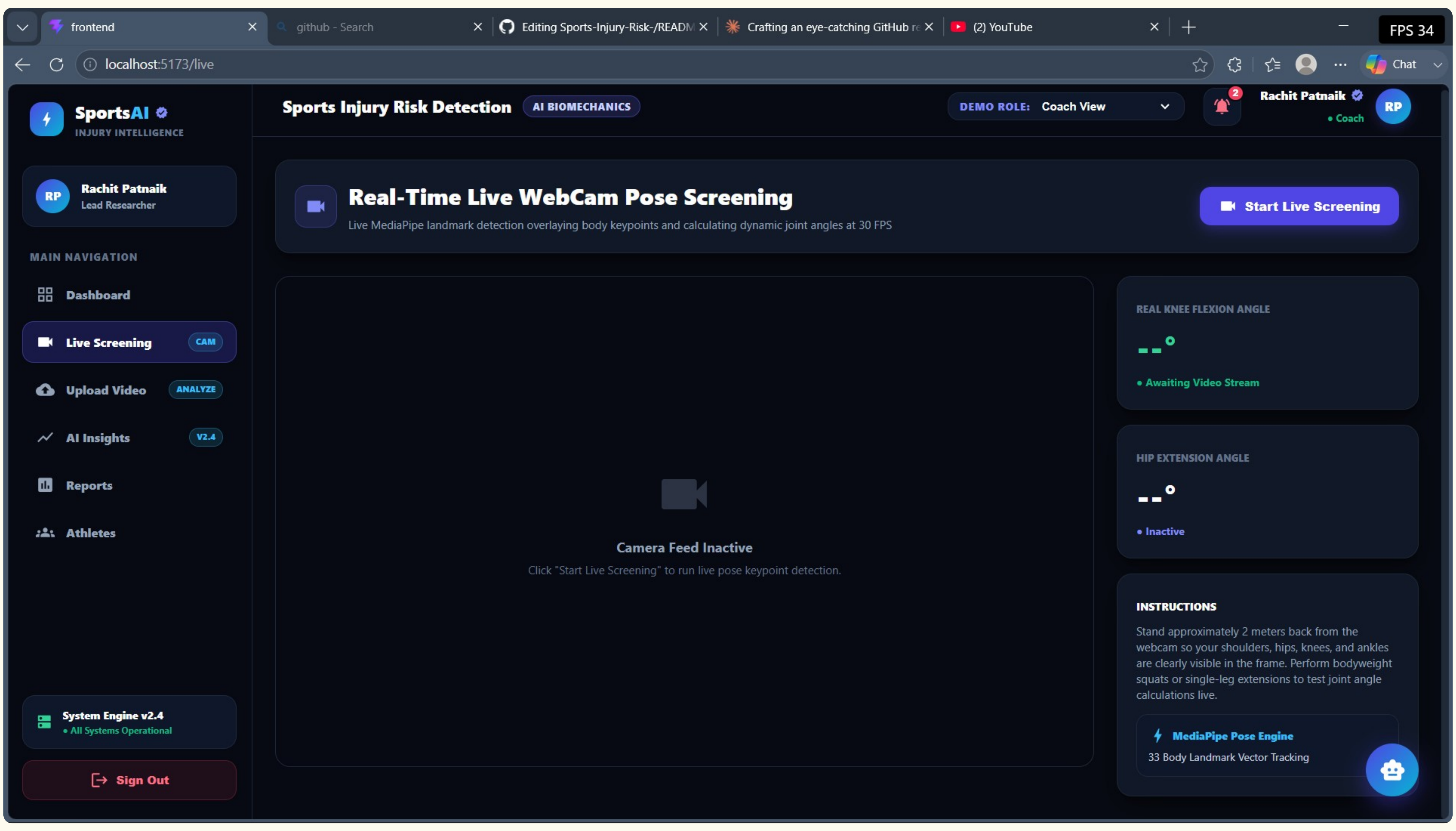


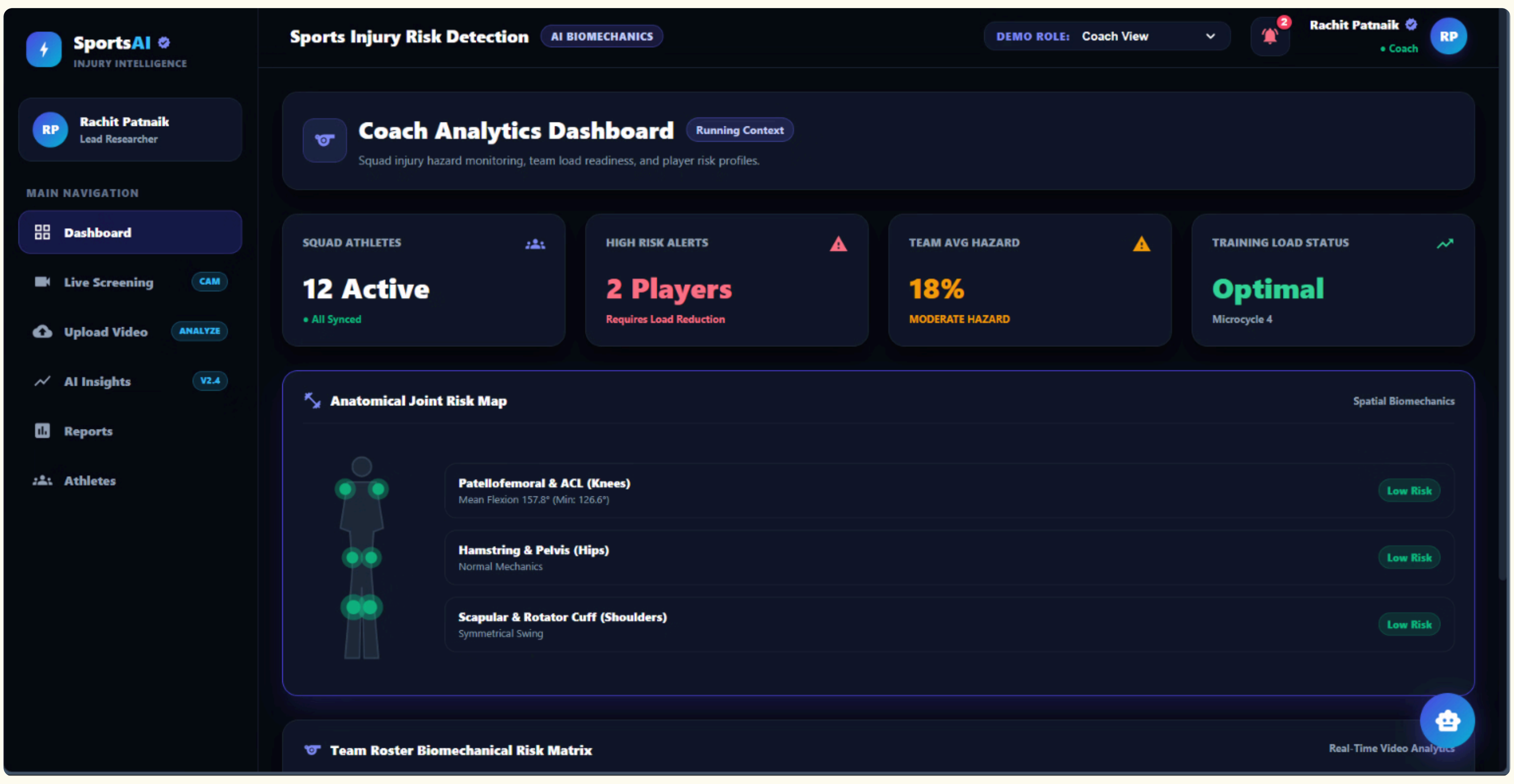
Kinematic Time-Series Insights

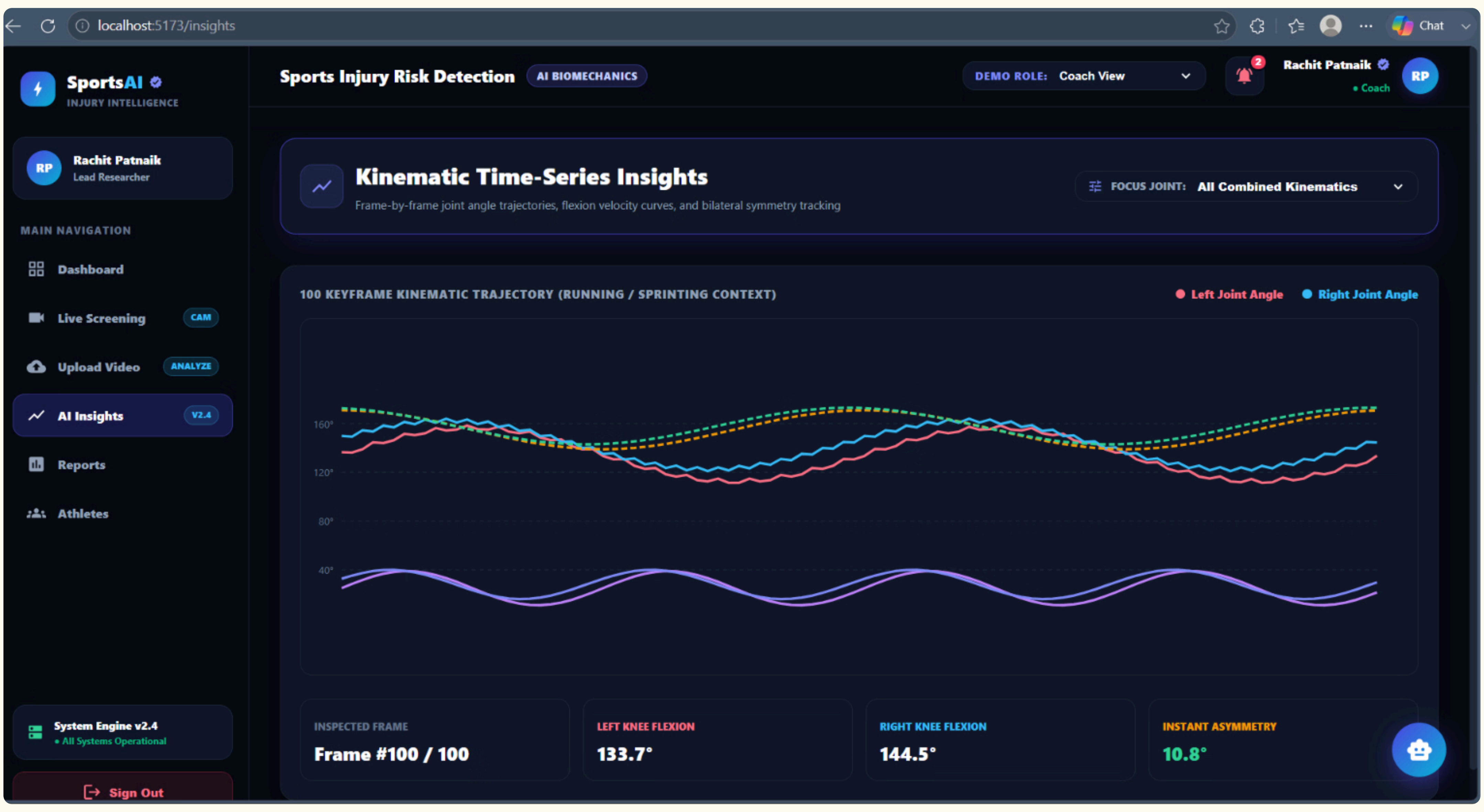
Time-series tracking highlights joint motion patterns, comparisons, and movement quality over time.

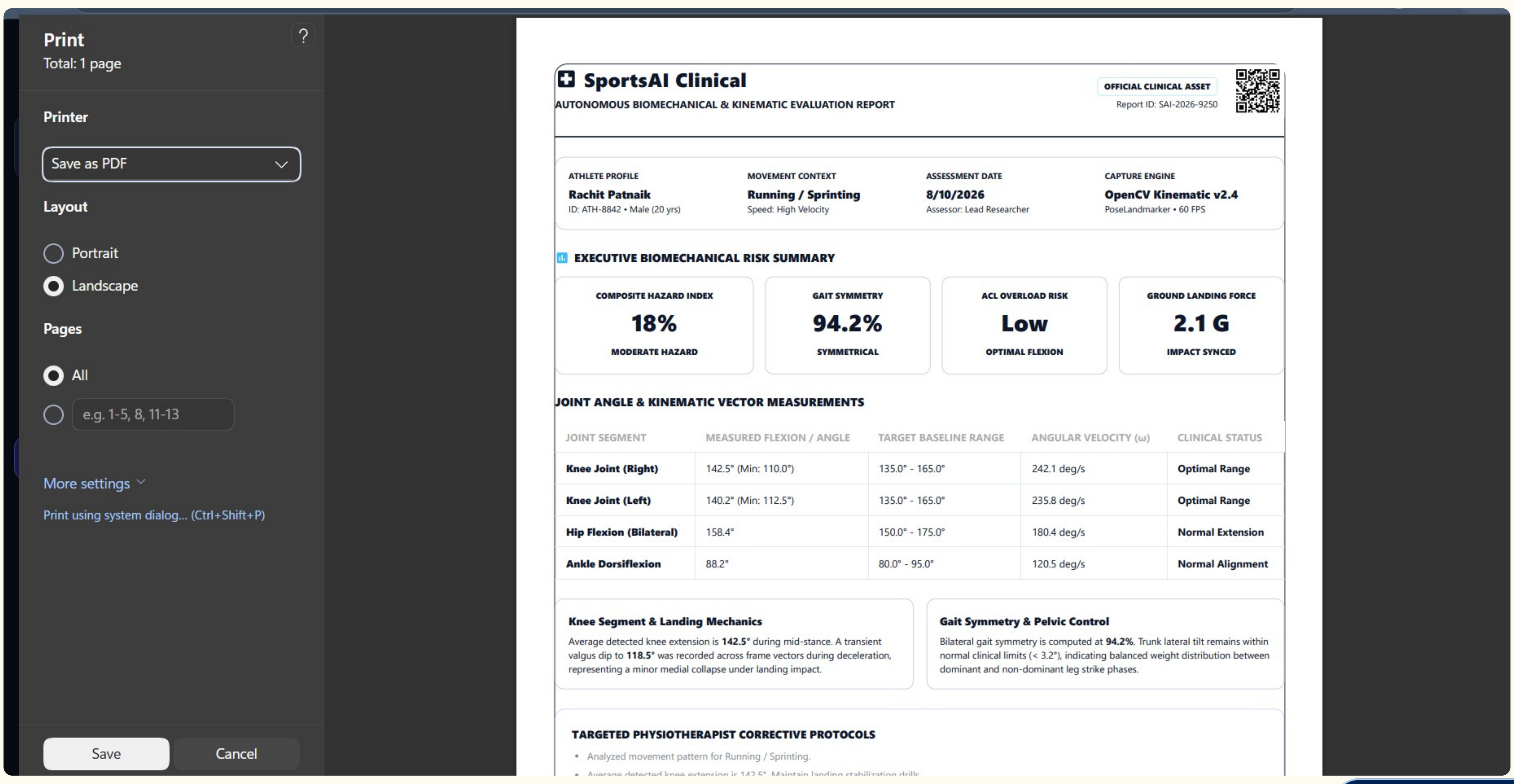
Clinical PDF Report

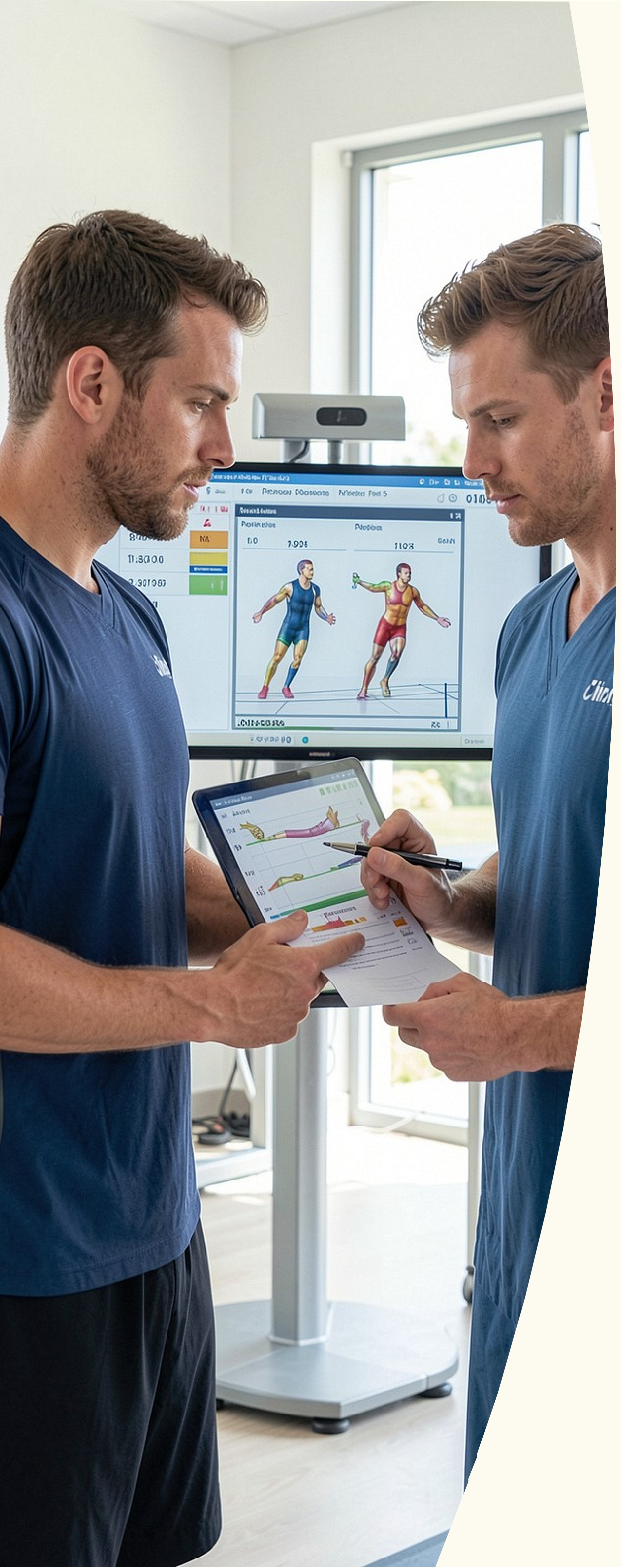
Exportable report format designed for sharing assessment results in a polished clinical document.











Measurable Outcomes and Positive Impact

Proactive Risk Detection

Identifies subtle, high-risk movements before acute injury occurs — shifting care from reactive to preventive.

Reduced Manual Coordination

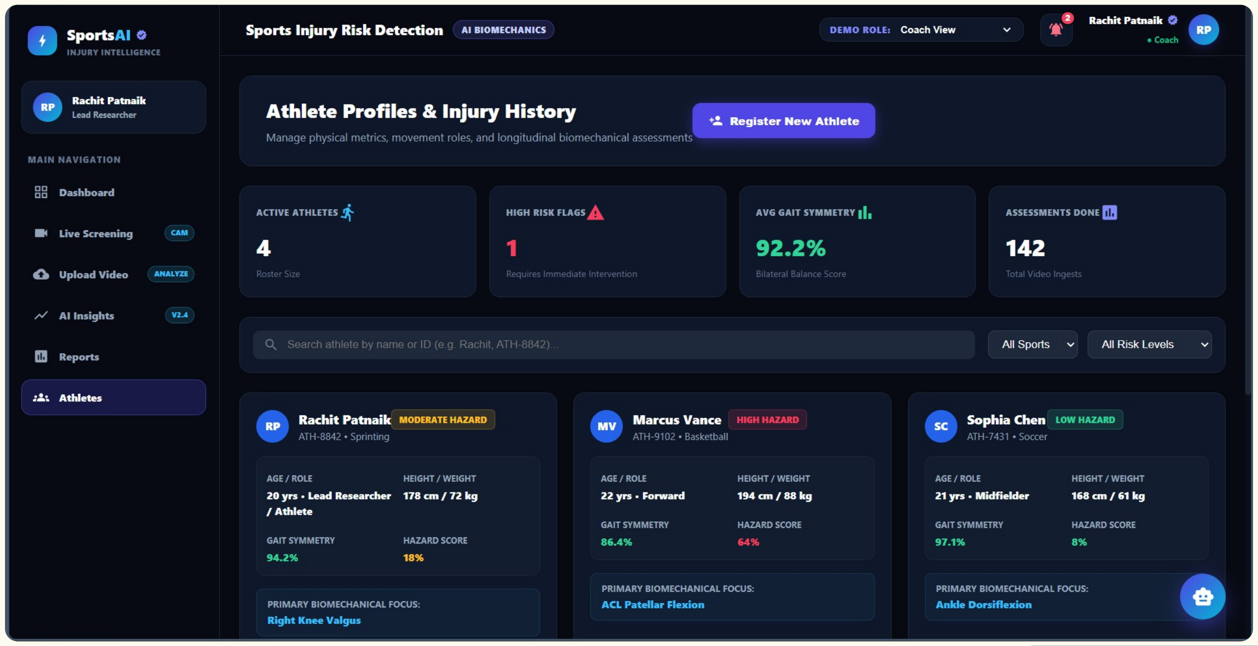
Automated generation minimizes hours spent on manual biomechanical analysis and reporting.

Enhanced Team Collaboration

Provides a unified, data-driven view of athlete health across all roles and disciplines.

Improved Athletic Performance

Optimizes movement quality through targeted corrective protocols and personalized guidance.



Future Enhancements

Physical Wearable Integration

Syncing live hardware IMUs directly with vision data for richer, multi-modal biomechanical insight.

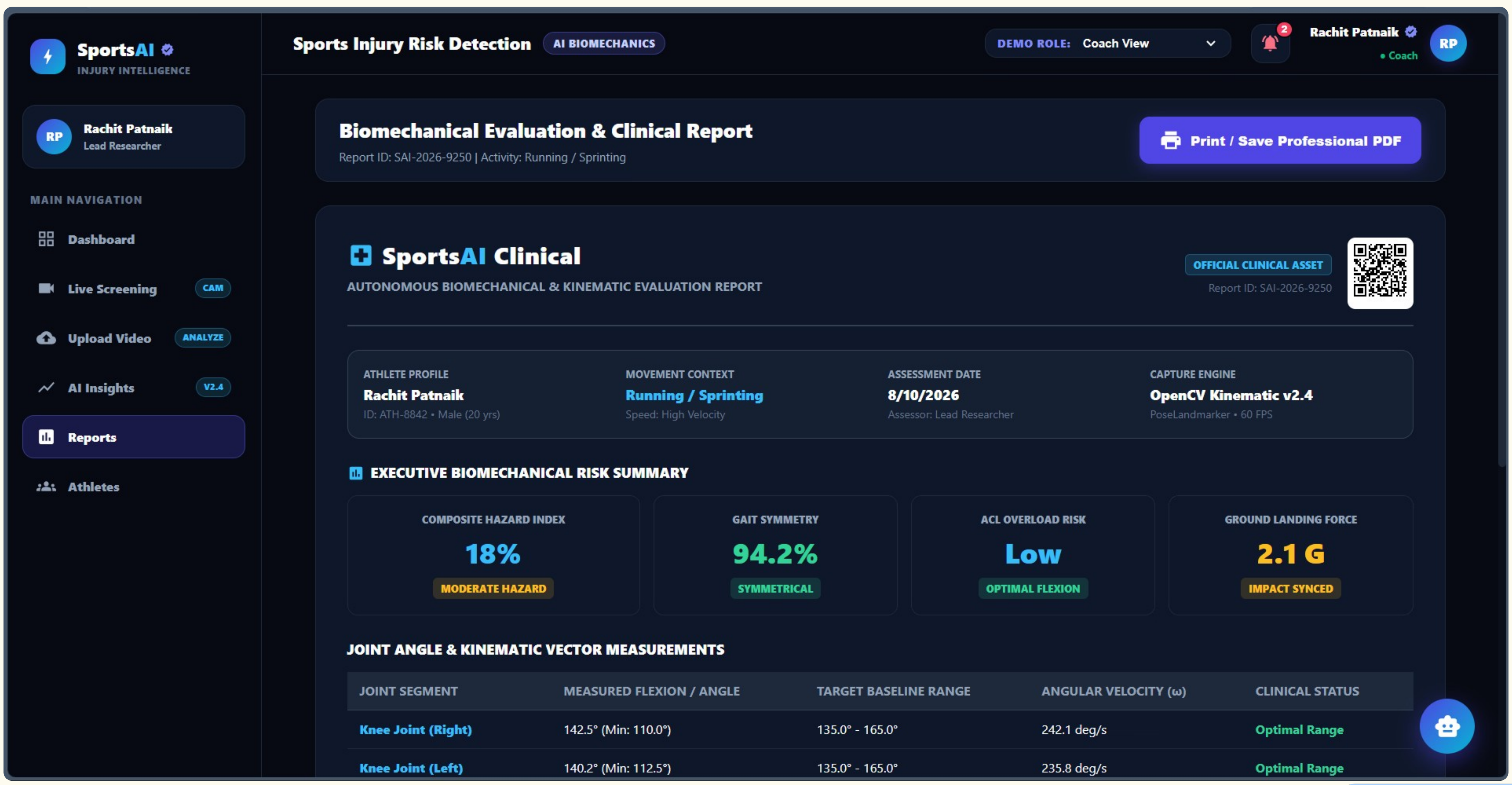
Advanced 3D Biomechanics

Upgrading from 2D planar angle calculation to full 3D spatial kinematics for greater accuracy.

Predictive Deep Learning Models

Training complex LSTM/CNN models on historical injury datasets to forecast risk with greater precision.

The automated clinical reports combine biomechanical analysis with QR verification, demonstrating the platform's ability to generate professional, actionable clinical documentation.



Thank You

Connect & Explore

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Role: B.Tech Computer Science & Engineering

Project Repository: github.com/springboardmentor1234r/Sports-Injury-Risk- (Branch: Rachit_Patnaik)

Interactive API/App: http://localhost:5173 | http://localhost:8000/docs

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